Supporting Document: Living Masterplan

Subtitle: A Deterministic Township Decision Engine for Regenerative Development
Hackathon: Sime Darby Property University Hackathon 2026
Theme: PLANET (Circular Economy & Climate Resilience)
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1. Executive Summary

Living Masterplan is not a generative chatbot. It is a deterministic asset performance and simulation engine wrapped in a conversational LLM interface.
Our core design principle is: **"Tools calculate the world. The LLM reasons about the world."**
While traditional LLMs suffer from hallucinations, our system anchors every financial, spatial, and environmental metric to rigorous mathematical models. The LLM is strictly restricted to orchestrating context, generating candidate plans, and reasoning over verified simulations.

2. Backend Architecture (The Pipeline)

The engine processes data through four deterministic layers before the LLM is invoked:

- **B3 (Derived Metrics)**: Computes spatial accessibility, carbon footprints, and demographic demands.
- **B6 (Deadspace Detection)**: A three-layer rule-based diagnostic engine that flags underperforming assets.
- **B4 (Simulation Engine)**: A multi-year projection model (2026-2033) simulating the impact of interventions.
- **B7 (LLM Orchestrator)**: Parses intent, generates candidates, runs simulations, and selects the optimal plan via post-simulation reasoning.

3. Core Deterministic Models & Scientific Basis

3.1 Spatial Topology & Network Accessibility

- **Door-to-Door Routing**: We use **Dijkstra's shortest path algorithm** on a pre-processed walking/driving graph. Off-network distances (from building centroid to the nearest road edge) are added to the graph traversal to calculate true door-to-door accessibility.

- **Component Bridging**: OSM clipped networks are often fragmented. We built a bridge_components algorithm using KD-Trees and a **1.35 urban detour factor** to connect isolated network islands, ensuring accurate walkability.
- **Huff Gravity Model**: Commercial demand allocation is calculated per-subtype. A supermarket competes in a 3km radius, while a cafe competes in an 800m radius. We use an attractiveness parameter (area) and a distance decay factor (**Beta = 2.0**) to determine market share.

3.2 Subtype-Specific Carbon Accounting & Environmental Science

We do not apply a flat carbon multiplier. Every asset's footprint is calculated based on its specific sub-business type (Subtype):

- **Operational Carbon (Energy)**: Each subtype has a specific energy intensity benchmark (kWh/m²/year) defined in our taxonomy.yml. For example, a supermarket (high refrigeration load) is modeled at **420 kWh/m²**, while a standard office is **210 kWh/m²**. We apply a vertical penalty (×1.15) for offices over 10 storeys. The grid emission factor used is **0.769 kgCO2e/kWh** (Source: Suruhanjaya Tenaga / Energy Commission Malaysia).
- **Embodied Carbon (Life-Cycle)**: We use localized material intensities (tCO2e/m²) and structural lifespans per subtype. A concrete terrace house (60-year lifespan) has an intensity of **0.28**, while a long-span steel supermarket (30-year lifespan) is **0.42**. These are based on the **CIDB (Construction Industry Development Board) Malaysia** embodied carbon inventory. Retaining the structural shell (adaptive reuse) reduces this intensity to just **0.08**.
- **Urban Heat Island (UHI) Proxy**: Calculated via a microclimate formula: $\Delta T = 8.0 * (1 - canopy_coverage) - 4.5 * canopy_coverage$. We also model **Canopy Growth** geometrically (2.5% annual growth, capping at 60% site coverage).

3.3 Occupancy Dynamics & Financial Modeling

- **Damped Occupancy Trajectory**: Occupancy doesn't jump instantly. We use a damped adjustment formula: $O_t = O_{t-1} + \lambda(O_t^* - O_{t-1}) - \kappa \cdot \text{tipping_point_gap}$
- . This simulates realistic market absorption rates and death spirals.
- **Lease Schedule Awareness**: Future occupancy is bounded by physical lease expirations. We use a **75% renewal probability** per 3-year term, seeded deterministically per lease, to forecast if an asset will naturally empty out.
- **Financial Aging**: Maintenance costs compound at 3% annually (alpha = 0.03) while rental incomes scale linearly with occupancy. NOI is strictly derived from these updated parameters.

4. The Diagnostic Engine (B6: Deadspace Detection)

Instead of a single arbitrary "score," we use a medical-style triage system:

- **Layer 1 (Underperformance)**: Flags assets below benchmarks. Subtypes like cafes are evaluated on walking accessibility, while supermarkets are evaluated on driving. Includes stranded carbon and UHI diagnostics.
- **Layer 2 (Value Failure)**: Flags negative NOI, severe maintenance surges, or insufficient service coverage.
- **Layer 3 (Actionability & Authority)**: Checks legal and physical constraints. We strictly separate full_control (developer executes) from advisory (LLM issues recommendations only).
- **Trajectory Classification**: We analyze 5-year historical occupancy data to classify assets as rapid_decline, chronic_low, recovering, or never_alive.
- **Urgency Index**: Root-cause flags are grouped and weighted to prevent an asset with 10 correlated flags from unfairly dominating the list.

5. LLM Orchestration & Guardrails (B7)

To prevent the LLM from hallucinating, we enforce strict boundaries:

5.1 The Context Pack

The LLM is never given the whole database. context_builder.py dynamically slices the data (Target assets, nearby competitors, spatial rings, available legal interventions, and static assumptions) based on the user's intent.

5.2 The Validator (Anti-Hallucination)

We built a ProposalValidator that intercepts every LLM proposal. It physically checks:

- **Authority Boundaries**: If an asset is advisory, the LLM cannot execute a repurpose.
- **Budget Limits**: Cumulative CAPEX cannot exceed the user's stated budget.
- **Mixed-Ownership Constraints**: Warns if a repurpose targets an asset where units have already been sold (Strata/JMB).

5.3 Multi-Candidate Simulation & Catalytic Verification

1. The LLM generates 2-3 radically different plans.
2. The B4 simulation engine runs all of them simultaneously, producing a simulation_brief with actual NOI, carbon, and occupancy deltas.
3. If the LLM claims a "Catalytic Intervention" (e.g., building a sports center to benefit a nearby mall), we **verify this mathematically**. If the simulation shows the beneficiary's reachable population didn't increase, we automatically downgrade the

LLM's claim from verified to partially_verified or claimed.
(Based on: Ewing & Cervero 2010 — a 10% change in accessibility only yields a 2.5% change in actual use).

6. Key Assumptions & Future Limitations

- **Huff Attractiveness**: Currently proxied by lettable floor area. Future iterations should integrate actual tenant rent value or brand attractiveness.
- **Solar Generation**: Calculated based on tropical averages (1350 kWh/kWp/year) with a 60% usable land fraction. Actual yield requires on-site engineering surveys.
- **Demand Migration**: We assume population growth is uniform across the township. A micro-simulation of residential migration could improve localized accuracy.

7. System Architecture

